

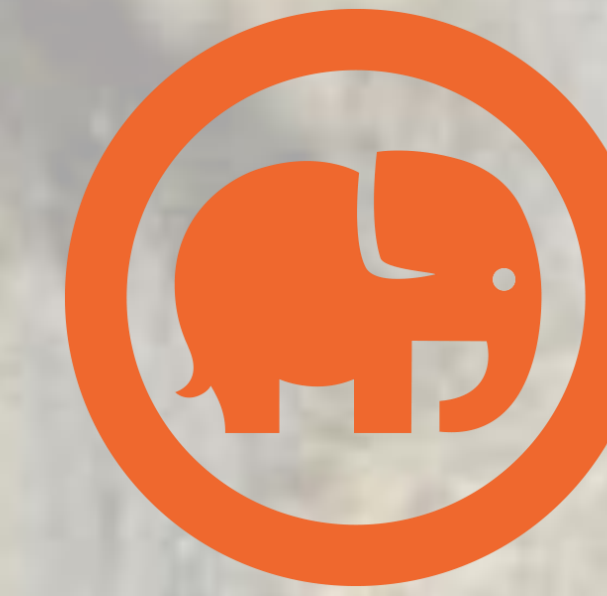
3D digitization of Yellowstone wolf skulls for education, research, and outreach



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Introduction

3D digitization of museum collections creates **digital archives** of physical objects and **increases accessibility** for research and education

Yellowstone's Heritage and Research Center houses one of the best-studied collections of wolf skulls in the world ($n > 225$) with associated metadata including age, sex, pack composition, social status, coat color, disease history, pedigree, and much more

E.g., Digitized Cranium of 302M

As of December 2022, we have 205 skulls photographed including that of 302M (shown below).



"For the benefit and enjoyment of the people" – wherever they are.
- **Yellowstone National Park Protection Act, 1872**

Implications

Specimen virtualization opens access to **experiential learning** for anyone with internet access. Educators can create interactive online experiences to teach students about animal diversity and evolution.

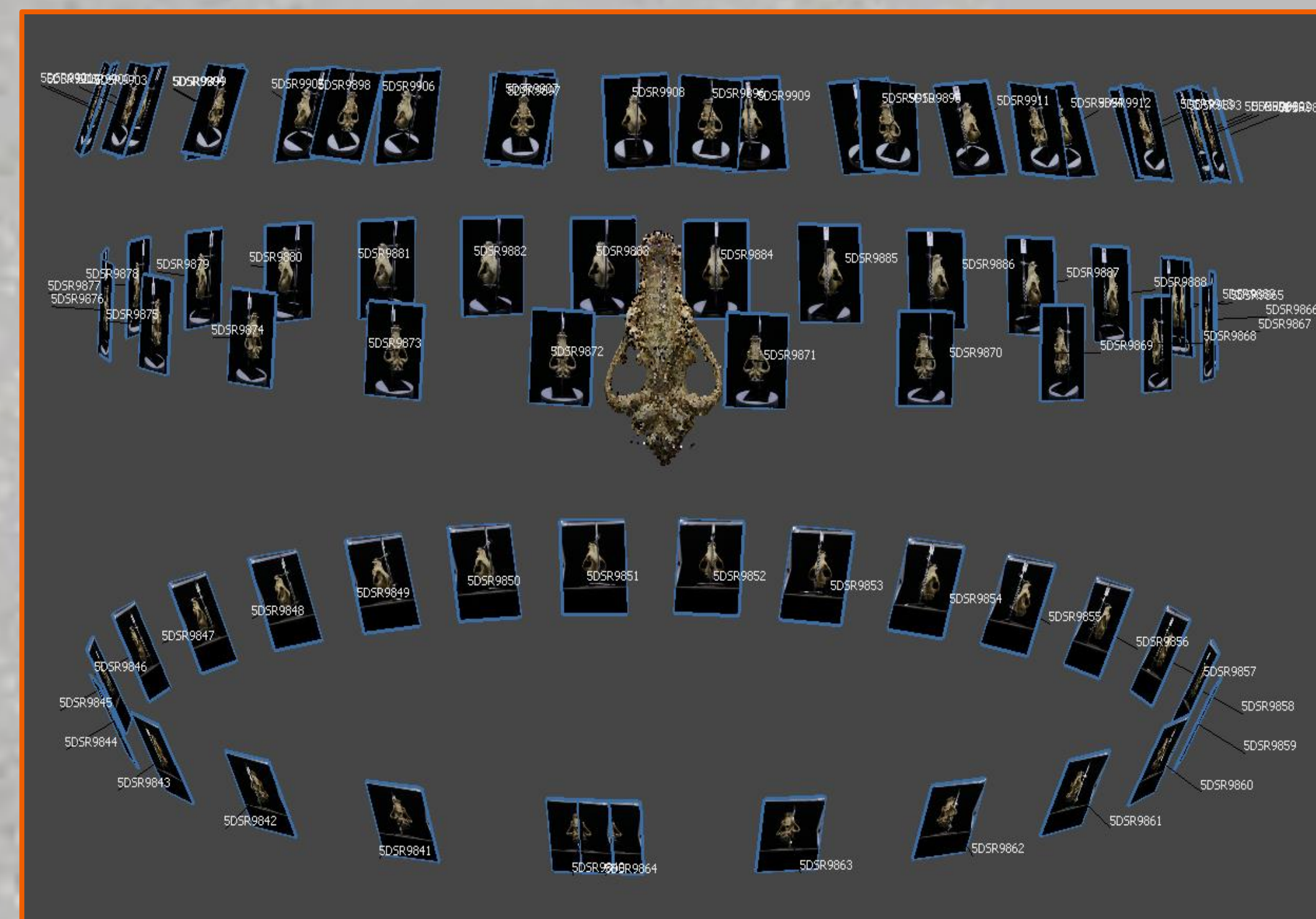
Researchers can use landmark-based **geometric morphometric methods** to study morphological and functional variation of skull shape. Also, this collection can be compared with other digitized skulls from around the world.

Acknowledgements

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Step 1: Create a photo image series of the cranium and mandible for each specimen.



Step 2: Create a 3D image using photogrammetry software. We are using Agisoft Metashape.



Step 3: Upload the 3D image to an online repository. We are using MorphoSource.

302M Digital Cranium



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